PLANRS ACADEMY

The Science of Wonder: Grade 3 Physics

Static Electricity

The Dancing Cereal Experiment

Session 1: The Invisible Pull

Are You a Science Magician?

Have you ever taken off your school sweater on a cold winter morning and felt a tiny 'tick-tick' sound? Or watched your hair try to fly away while combing? Today, we will learn how to control this invisible power to make cereal dance!

Your Science Mission

In this session, we will put on our magician's hats to achieve these goals:

- ⚡ **Understand Static Electricity:** Discover the "sticky power" made by rubbing.
- ⚡ **Master the Magic of Rubbing:** Learn how surfaces trade invisible charges.
- ⚡ **The Dancing Cereal Demo:** Use a balloon to make Muri (puffed rice) jump!
- ⚡ **Find Static at Home:** Spot everyday examples in winter clothes and blankets.

THE QUICK MYSTERY

Can you make hair stand up without touching it? Let's find out how!

What is Static Electricity?

Most electricity flows through wires to power our lights and fans. But Static Electricity is different—it likes to stay in one place on the surface of an object!

> **Definition:** Static Electricity is a build-up of electrical charge on the surface of an object, usually made by rubbing two things together.

The Magic of Rubbing

Everything in the world is made of tiny invisible pieces. When we rub two things together (like a balloon against a woollen sweater), these tiny pieces jump from one object to another. This leaves one object with a "sticky" electric charge!

Everyday Magic: Static Around Us

You don't need a high-tech laboratory to see static electricity. It happens in our Indian homes all the time:

🧣 **The Winter Sweater:**
When you take off your school sweater in December or January, you often hear crackling noises. In the dark, you might even see tiny blue sparks!

🦱 **The Plastic Comb:**
After combing dry hair quickly with a plastic comb, bring the comb near small bits of paper. Watch how the paper jumps up to hug the comb!

🥻 **Silk Sarees:**
Sometimes, a festive silk saree clings tightly to your legs or makes tiny snapping sounds when it is unfolded. That's static cling!

The Experiment: The Dancing Cereal

Let's use this invisible sticky force to create a science magic show using a common Indian snack!

What We Need:

🎈 An inflated balloon.
🎈 A handful of **Muri** (puffed rice / Murmura).
🎈 A dry woollen cloth, a woollen sweater, or your own dry hair!
🎈 A flat plate or table surface.

[Illustration: A balloon being rubbed on a sweater next to a plate of puffed rice grains]

The Science Show Steps

Follow these steps to make your cereal jump and dance:

Step 1: Spread a handful of Muri flat on a plate or a clean table.

Step 2: Hold the balloon over the Muri. Does anything happen? No, because the balloon has no charge yet!

Step 3: Now, rub the balloon quickly against your hair or a woollen sweater for 20-30 seconds. You are now charging the balloon!

Step 4: GENTLY bring the rubbed side of the balloon close to the Muri without letting it touch the plate.

💥 Snap! Crackle! Jump! 💥

Watch in wonder as the pieces of Muri fly off the plate and stick to the balloon like little acrobats!

Why Does the Muri Dance?

Let's unlock the science secret behind our magic show. It boils down to two main actions:

1. Charging by Friction

When you rub the balloon on your hair or sweater, the friction forces tiny negative charges called electrons to move from your hair onto the rubber surface of the balloon. The balloon now has a strong negative static charge.

2. Electrical Attraction

Even though the pieces of Muri don't have a charge, the strong negative charge on the balloon acts like an invisible hook. It pulls on the opposite charges inside the light pieces of Muri, lifting them right off the plate!

> 💡 **EXPERIMENT TWIST: After a few seconds, some pieces of Muri will suddenly shoot AWAY from the balloon! This happens because they steal some charge from the balloon and get pushed away!**

Static Tricks to Try at Home

Now that you are a certified Science Magician, try these other two-minute tricks:

💧 1. THE BENDING WATER TRICK

Turn on a kitchen tap so that a very thin, smooth stream of water flows down. Charge your balloon by rubbing it on your hair, then hold it close to the stream of water. Watch the water bend towards the balloon!

📄 2. THE PAPER PICKER

Tear up tiny bits of tissue paper. Charge your balloon and see how many bits you can pick up at once.

🧱 3. THE BALLOON WALL-STICK

Charge your balloon and try to stick it to the wall. How long does it stay there before the static charge wears off?

The Science Safety Guide

Static electricity is fun, but we must use it wisely.

🛡️ **SAFETY FIRST!**

• **Tiny Shocks:** Static sparks might give you a tiny surprise (like a pin-prick), but they are not dangerous. However, never play with real electrical sockets or wires!

• **Balloon Care:** Don't rub the balloon so hard that it pops. Pop sounds can be scary for pets!

• **Clean Up:** Always put your Muri back in the bin or give it to the birds after your experiment.

Interactive Challenge: Track Your Sparks

Fill out this mini lab-report after testing your powers!

1. My Balloon rubbed best against: [Hair / Sweater / Cotton Towel]

2. How many pieces of Muri jumped up? ___________ pieces.

3. Did the balloon stick to the wall? [Yes / No]

4. If yes, for how many minutes? ___________ minutes.

📝 **Teacher's Note:** This experiment works best on dry days! If it is very humid or raining outside, moisture in the air will steal the electric charges before the magic happens.

Lesson Recap

You have successfully completed your first session as a Science Magician! Let's remember the big ideas:

🌟 **Static Electricity** is built-up charge on a surface.
🌟 **Rubbing** is the key to creating this "Sticky Power".
🌟 It can **Pull** light objects like cereal or paper without touching them.
🌟 We see it in sweaters, combs, and silk sarees.

Stay Curious!

The world is full of invisible forces waiting to be explored. See you in the next science session!